# Grade 11 – Term 1
# C2 Practice Activity 2 with Solutions

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## Evaluated criterion

- C2: Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.

## 1 Introduction

**Problems:**


1.1. Equation of a line through two points

1.2. Uniform distribution

1.3. Linear distribution

1.4. Triangular distribution

1.5. Piecewise uniform distribution

1.6. Piecewise linear distribution


The central skill in this activity is constructing a continuous probability model from a situation, rather than merely evaluating a supplied formula. For every problem, use the same six-step method: (1) identify the random variable, its **support**, breakpoints, endpoint heights, and geometric shape; (2) use total area = 1 to determine all unknown heights or constants; (3) apply the straight-line rule to derive every sloping formula (or note that no line equation is needed for constant sections); (4) write the complete pdf, including 0 outside its support; (5) identify the requested tail shape

and calculate its area; and (6) identify the shapes in the requested interval or band and calculate their areas, using a complement when helpful. Thus Steps 1–4 construct the pdf, Step 5 calculates the requested tail, and Step 6 calculates the requested interval or central band. State each result in context when useful. A density is a height, so it may exceed 1; it is the *area* that must total 1. Endpoints do not affect a continuous probability, for example $P(a < X < b) = P(a \le X \le b)$.

## 1.1 Equation of a line through two points

Two distinct points determine exactly one straight line. Suppose the known points are given. This method will later be used to construct the formulas for linear and piecewise-linear probability density functions from known points on their straight segments. Write the two points as

$$(x_1, y_1), \qquad (x_2, y_2).$$

When $x_1 \neq x_2$, the first step is to calculate the slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

The slope measures the change in the vertical coordinate for each one-unit change in the horizontal coordinate. The subtractions must use the same order in the numerator and denominator: moving from point 1 to point 2 gives $y_2 - y_1$ and $x_2 - x_1$. Reversing both subtractions gives the same slope, but reversing only one changes its sign incorrectly.

After finding $m$, use either known point in point-slope form. Using the first point gives

$$y - y_1 = m(x - x_1).$$

Here $(x, y)$ represents any point on the line. Expanding and isolating $y$ produces slope-intercept form:

$$\begin{aligned} y - y_1 &= mx - mx_1, \\ y &= mx + (y_1 - mx_1) = mx + b. \end{aligned}$$

Thus, once the slope is known, the intercept can also be calculated directly from

$$b = y_1 - mx_1.$$

Either known point may be used in point-slope form or in the intercept formula; both must produce the same line equation.

If $x_1 = x_2$, the denominator in the slope calculation is zero. The line is vertical, its equation is $x = x_1$, and the usual slope-intercept procedure does not apply.

**Worked example.** Find the line through

$$(2, \tfrac{1}{6}) \qquad \text{and} \qquad (6, \tfrac{1}{3}).$$

These coordinates give

$$x_1 = 2, \quad y_1 = \frac{1}{6}, \qquad x_2 = 6, \quad y_2 = \frac{1}{3}.$$

First calculate the slope:

$$m = \frac{\frac{1}{3} - \frac{1}{6}}{6 - 2} = \frac{\frac{1}{6}}{4} = \frac{1}{24}.$$

Substitute the first point and this slope into point-slope form:

$$y - \frac{1}{6} = \frac{1}{24}(x - 2).$$

Expanding and isolating $y$ gives

$$y - \frac{1}{6} = \frac{x}{24} - \frac{2}{24},$$
$$y = \frac{x}{24} - \frac{1}{12} + \frac{1}{6} = \frac{x}{24} + \frac{1}{12}.$$

Equivalently, the intercept formula gives

$$b = y_1 - mx_1 = \frac{1}{6} - \frac{1}{24}(2) = \frac{1}{6} - \frac{1}{12} = \frac{1}{12}.$$

Therefore the final equation is

$$y = \frac{x}{24} + \frac{1}{12}.$$

Verify both original points. For $(2, 1/6)$,

$$\frac{2}{24} + \frac{1}{12} = \frac{1}{12} + \frac{1}{12} = \frac{1}{6},$$

and for $(6, 1/3)$,

$$\frac{6}{24} + \frac{1}{12} = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}.$$

Both points satisfy the final equation. In later sections, the two coordinates may represent an input value and the corresponding height of a linear density.

**Algorithm: line through two points.**

1. Identify $(x_1, y_1)$ and $(x_2, y_2)$.
2. Compute $m = \dfrac{y_2 - y_1}{x_2 - x_1}$.
3. Compute $b = y_1 - mx_1$.
4. Write $y = mx + b$.
5. Check that both given points satisfy the equation.



## 1.2 Uniform distribution

A uniform density has one constant height on support $[L, U]$:

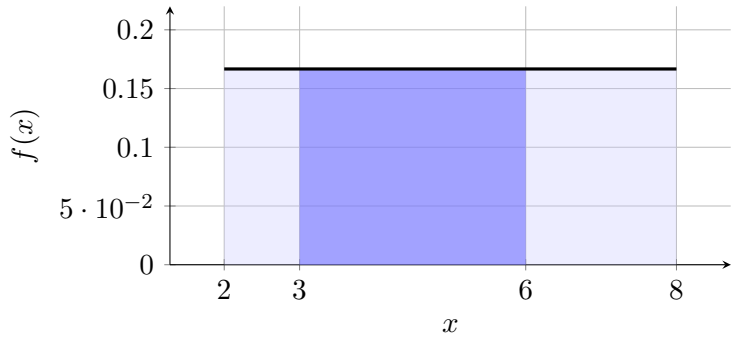
$$f_X(x) = \begin{cases} h, & L \le x \le U, \\ 0, & \text{otherwise.} \end{cases} \qquad (U - L)h = 1, \quad h = \frac{1}{U - L}.$$

Thus every probability is a rectangle: $P(a \le X \le b) = (b - a)h$ for $L \le a < b \le U$.

**Worked example.** A delivery time $T$ is equally plausible from 2 to 8 minutes. The support is $[2, 8]$ and the whole rectangle has width 6 and height $h$, so $6h = 1$ and $h = 1/6$. For $3 \le T \le 6$, the relevant rectangle has width $6 - 3 = 3$ and height $1/6$:

$$P(3 \le T \le 6) = A_{\text{rectangle}} = 3\left(\frac{1}{6}\right) = \frac{1}{2}.$$

There is a 50% chance that delivery takes from 3 to 6 minutes.




## 1.3 Linear distribution

A single linear density has $f_X(x) = mx + d$ throughout one support $[L, U]$ (and 0 outside), with nonnegative endpoint heights $h_L = f(L)$ and $h_U = f(U)$. Its whole area is the trapezoid

$$1 = \frac{1}{2}(h_L + h_U)(U - L).$$

An interval $[a, b]$ is also a trapezoid of width $b - a$ and parallel heights $f(a), f(b)$. A triangle is only the special case in which one endpoint height is zero.

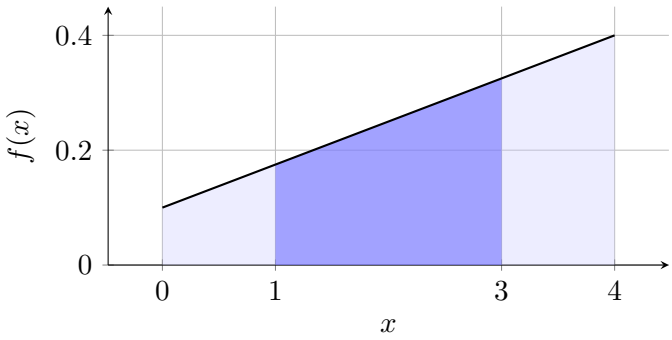
**Worked example.** A shop's daily price adjustment $X$ (in dollars) lies between 0 and 4. Its likelihood density increases linearly, and the density at 4 is four times that at 0. Translating the description, let the endpoint heights be $h$ and $4h$. The whole trapezoid gives

$$1 = \tfrac{1}{2}(h + 4h)4 = 10h, \qquad h = 0.1.$$

The line through $(0, 0.1)$ and $(4, 0.4)$ is $f_X(x) = 0.075x + 0.1$ on the support (and 0 otherwise). For $1 \le X \le 3$, $f(1) = 0.175$ and $f(3) = 0.325$. The required figure is a trapezoid of width 2 with those parallel heights:

$$P(1 \le X \le 3) = \tfrac{1}{2}(0.175 + 0.325)(2) = 0.5.$$

Both support endpoints have positive density, so this is genuinely linear, not triangular.




## 1.4 Triangular distribution

A triangular density has support $[L, U]$, height 0 at both ends, and mode $M$ ($L < M < U$) with height $H$. It rises linearly on $[L, M]$ and falls linearly on $[M, U]$. Since the whole figure has base $U - L$,

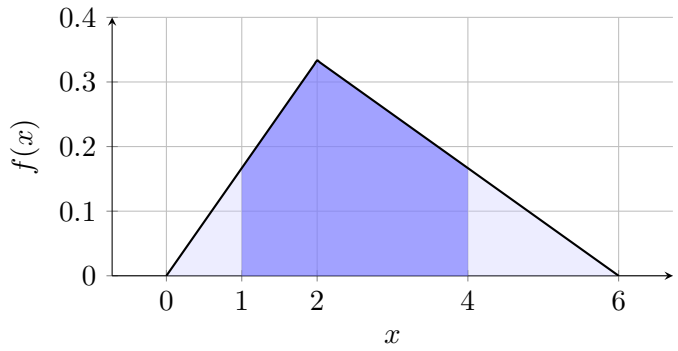
$$1 = \tfrac{1}{2}(U - L)H, \qquad H = \frac{2}{U - L}.$$

Heights inside either sloping side are found by proportional similar triangles.

**Worked example.** A project's cost overrun $C$ (thousand dollars) has minimum 0, most likely value 2, and maximum 6. Its support is $[0, 6]$, mode is 2, and $\frac{1}{2}(6)H = 1$, so $H = 1/3$. To find $P(1 \le C \le 4)$, subtract two tail triangles. At $C = 1$ the height is $1/6$; at $C = 4$ it is also $1/6$. The left tail has base 1, height $1/6$; the right tail has base 2, height $1/6$:

$$P(1 \le C \le 4) = 1 - \tfrac{1}{2}(1)(\tfrac{1}{6}) - \tfrac{1}{2}(2)(\tfrac{1}{6}) = 1 - \tfrac{1}{12} - \tfrac{1}{6} = \tfrac{3}{4}.$$

The region crosses the mode, but subtraction avoids splitting it into two trapezoids.



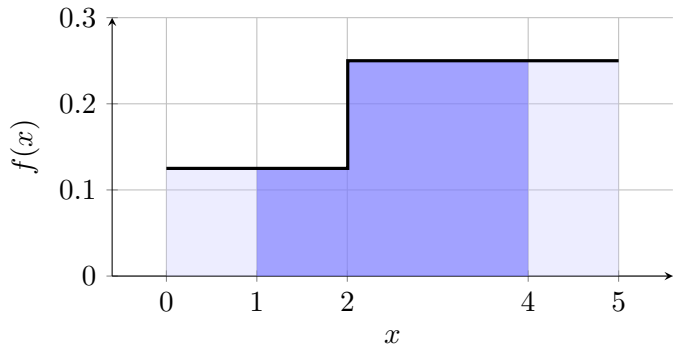



## 1.5 Piecewise uniform distribution

Here the density is constant *within* each interval but may jump at a breakpoint. Normalisation is the sum of rectangles: $1 = \sum(\text{segment width})(\text{segment height})$.

**Worked example.** A warehouse waiting time $W$ ranges from 0 to 5 minutes. Waiting times are equally dense within each of two regimes, 0–2 and 2–5, but the later regime has twice the density of the earlier one. If the first height is $k$, the narrative makes the second $2k$. A width-2, height-$k$ rectangle plus a width-3, height-$2k$ rectangle must have area 1: $2k + 6k = 1$, so $k = 1/8$. Thus the pdf has heights $1/8$ and $1/4$ on those respective regimes, and is 0 otherwise. The event $1 \le W \le 4$ spans the jump and uses two rectangles:

$$P(1 \le W \le 4) = \underbrace{(2-1)(1/8)}_{A_1} + \underbrace{(4-2)(1/4)}_{A_2} = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}.$$






### 1.6 Piecewise linear distribution

A piecewise-linear density joins straight segments at stated breakpoints. Read the height at each endpoint and breakpoint, then treat each piece as a triangle, rectangle, or trapezoid. Continuity means adjacent pieces meet at the same height.

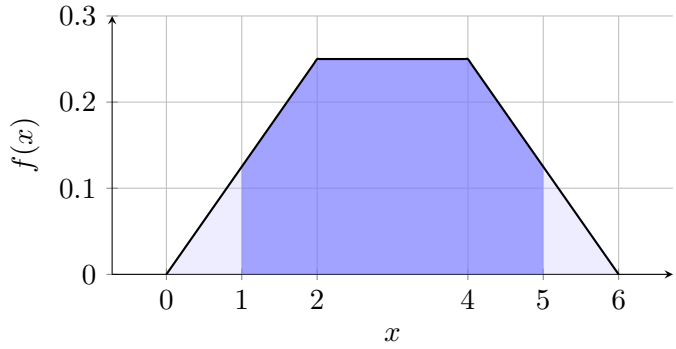
**Worked example.** Demand $D$ ranges from 0 to 6. Its density starts at zero, rises linearly to its maximum at 2, stays at that maximum until 4, and then falls linearly to zero at 6. Calling the maximum $H$ translates the story into a left triangle (base 2), rectangle (width 2), and right triangle (base 2):

$$1 = \tfrac{1}{2}(2)H + (2)H + \tfrac{1}{2}(2)H = 4H, \qquad H = \tfrac{1}{4}.$$

For $1 \le D \le 5$, subtract the two congruent tail triangles. By linearity, the heights at 1 and 5 are $H/2 = 1/8$:

$$P(1 \le D \le 5) = 1 - \tfrac{1}{2}(1)(\tfrac{1}{8}) - \tfrac{1}{2}(1)(\tfrac{1}{8}) = \frac{7}{8}.$$

This reusable approach identifies all three geometric pieces and uses subtraction from total area 1.





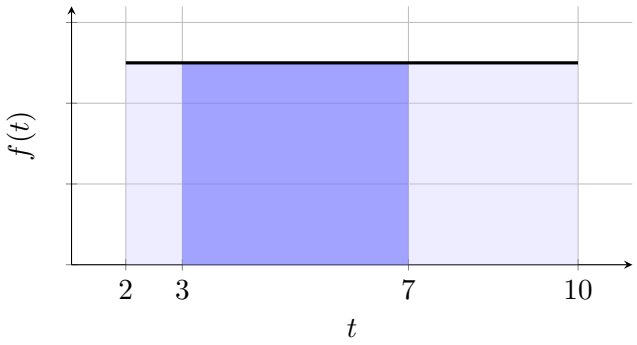

## 2 Uniform Distribution

**Problems:**

2.1. Microcredit disbursement window
2.2. Cold-chain loading time
2.3. Port inspection release time

### 2.1 Microcredit disbursement window

A Bogotá fintech releases approved working-capital loans after fraud and bank-account checks. The elapsed time can be measured to any fraction of an hour. Automatic cases never finish before 2 hours or after 10 hours, and operational records give no reason to favor one time over another anywhere between those limits. A bakery owner needs the money within 7 hours to meet a supplier's dispatch deadline. The fintech separately studies the 3–7 hour band when assigning automated-processing capacity.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $T$ from the narrative. Show how total area 1 determines every unknown value, give the required formula, and include 0 outside the possible range.

**C2.2** Calculate the deadline-risk tail probability $P(T > 7)$ geometrically.

**C2.3** Calculate the central-band probability $P(3 \leq T \leq 7)$ geometrically.

### C2.1 - Construct the pdf from the situation.

#### Step 1: Support and shape

The narrative defines $T$ as disbursement time, bounds it by 2 and 10, and says all times in that window are equally plausible. Thus the support is $[2, 10]$ and its density is a rectangle of unknown height $h$.

#### Step 2: Total area $= 1$

Normalisation gives

$$1 = A_{\text{rectangle}} = (10 - 2)h = 8h, \qquad h = \frac{1}{8}.$$

#### Step 3: Straight-line rule

The density is constant on each relevant section, so no line equation is required.

Therefore the complete model is

#### Step 4: Complete pdf

$$f_T(t) = \begin{cases} \frac{1}{8}, & 2 \leq t \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Its area is $8(1/8) = 1$, as required.

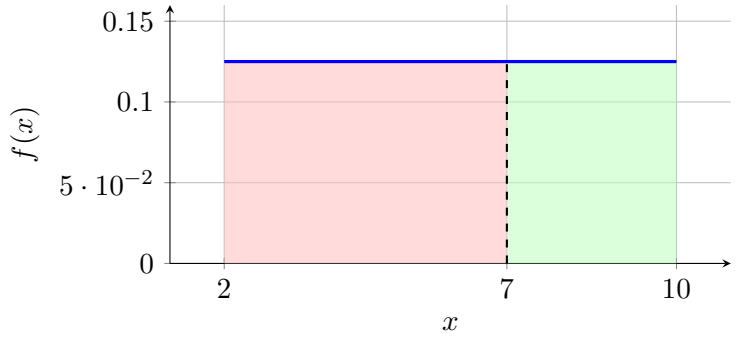
### C2.2 - Calculate the tail probability $P(T > 7)$.

#### Step 5: Tail probability

The right tail is a rectangle of width $10 - 7 = 3$ and height $1/8$:

$$P(T > 7) = A_{\text{rectangle}} = (10 - 7)\frac{1}{8} = \frac{3}{8} = 37.5\%.$$

Thus the bakery has a 37.5% modeled risk of missing its deadline.




**C2.3 - Calculate the interval probability $P(3 \le T \le 7)$.**

**Step 6: Interval probability**

The central band is a rectangle of width $7 - 3 = 4$:

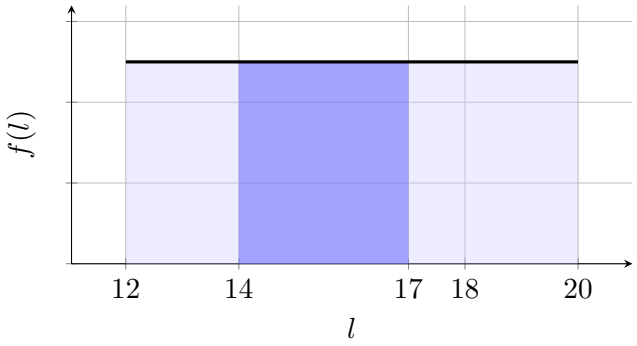
$$P(3 \le T \le 7) = (7-3)\frac{1}{8} = \frac{1}{2}.$$

Hence half of comparable disbursements occupy this capacity-planning band.



## 2.2 Cold-chain loading time

A flower exporter at El Dorado measures how long refrigerated pallets spend at the loading dock. A loading time may be recorded to any fraction of a minute, with the quickest possible loading taking 12 minutes and the slowest taking 20 minutes. Past operations make every time between those limits equally plausible. Taking more than 18 minutes risks a missed departure slot, while the 14–17 minute band is the target used to plan dock staffing.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $L$ from the narrative. Show how total area 1 determines every unknown value, give the required formula, and include 0 outside the possible range.

**C2.2** Calculate the missed-slot tail probability $P(L > 18)$ geometrically.

**C2.3** Calculate the central-band probability $P(14 \le L \le 17)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $L$ denote loading time in minutes. The narrative gives support $[12, 20]$ and equal plausibility throughout, so the graph is a rectangle of height $h$.

**Step 2: Total area** $= 1$

Normalisation gives

$$1 = (20 - 12)h = 8h, \qquad h = \frac{1}{8}.$$

**Step 3: Straight-line rule**

The density is constant on each relevant section, so no line equation is required.

Thus the complete pdf is

**Step 4: Complete pdf**

$$f_L(l) = \begin{cases} \frac{1}{8}, & 12 \le l \le 20, \\ 0, & \text{otherwise.} \end{cases}$$

Its rectangular area is $8(1/8) = 1$.

**C2.2 - Calculate the tail probability** $P(L > 18)$**.**

**Step 5: Tail probability**

The missed-slot tail is a rectangle of width $20 - 18 = 2$ and height $1/8$:

$$P(L > 18) = (20 - 18)\frac{1}{8} = \frac{1}{4} = 25\%.$$

**C2.3 - Calculate the central-band probability** $P(14 \le L \le 17)$**.**

**Step 6: Interval probability**

The target band is a rectangle of width $17 - 14 = 3$:

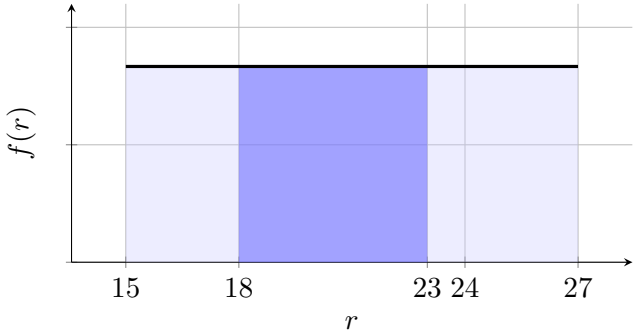
$$P(14 \le L \le 17) = (17 - 14)\frac{1}{8} = \frac{3}{8} = 37.5\%.$$

Thus 37.5% of modeled loadings fall in the staffing target.



## 2.3 Port inspection release time

An importer tracks the time from a container's arrival at Cartagena to its release after a routine inspection. The elapsed time can be recorded to any fraction of an hour, never falls below 15 hours, and never exceeds 27 hours. For routine inspections, every time between those limits is equally plausible. A release after 24 hours causes an extra storage charge. The importer uses the 18–23 hour band for its normal drayage schedule.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $R$ from the narrative. Show how total area 1 determines every unknown value, give the required formula, and include 0 outside the possible range.

**C2.2** Calculate the storage-charge tail probability $P(R > 24)$ geometrically.

**C2.3** Calculate the central-band probability $P(18 \leq R \leq 23)$ geometrically.

### C2.1 - Construct the pdf from the situation.

**Step 1: Support and shape**

Let $R$ denote release time in hours. The possible range is $[15, 27]$ and all times in it are equally plausible.

**Step 2: Total area** $= 1$

The rectangular height $h$ satisfies

$$1 = (27 - 15)h = 12h, \qquad h = \frac{1}{12}.$$

**Step 3: Straight-line rule**

The density is constant on each relevant section, so no line equation is required.

Therefore

**Step 4: Complete pdf**

$$f_R(r) = \begin{cases} \dfrac{1}{12}, & 15 \leq r \leq 27, \\ 0, & \text{otherwise.} \end{cases}$$

**C2.2 - Calculate the tail probability $P(R > 24)$.**

**Step 5: Tail probability**

The storage-charge tail is a width-3 rectangle:

$$P(R > 24) = (27 - 24)\frac{1}{12} = \frac{1}{4} = 25\%.$$

**C2.3 - Calculate the central-band probability $P(18 \le R \le 23)$.**

**Step 6: Interval probability**

The normal scheduling band has width $23 - 18 = 5$, so

$$P(18 \le R \le 23) = (23 - 18)\frac{1}{12} = \frac{5}{12} \approx 41.7\%.$$



# 3 Linear Distribution

**Problems:**

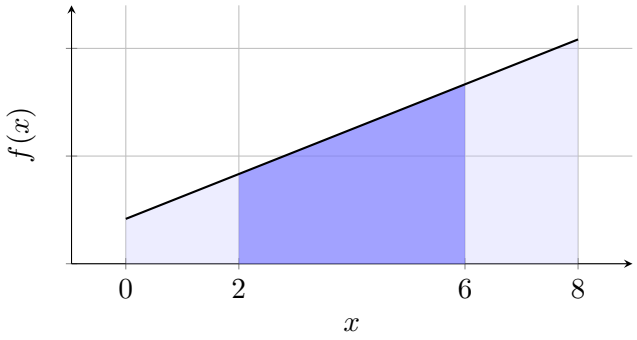

3.1. Coffee auction premium

3.2. Solar microgrid surplus

3.3. Customs-document correction time


## 3.1 Coffee auction premium

A Bogotá specialty-coffee roaster buys small lots from Colombian producers. The premium above the reference price can be measured to any fraction of a thousand pesos per kilogram. Auction rules keep it between 0 and 8 thousand pesos. As the premium grows, outcomes become steadily more common. Among two equally narrow price bands, one placed beside the upper limit occurs five times as often as one placed beside the lower limit. Premiums above 6 thousand pesos threaten margins, while the 2–6 thousand peso band can be absorbed without changing menu prices.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $X$ from the narrative. Show how total area 1 determines every unknown value, give the required formula, and include 0 outside the possible range.

**C2.2** Calculate the margin-risk tail probability $P(X > 6)$ geometrically.

**C2.3** Calculate the central-band probability $P(2 \le X \le 6)$ geometrically.

### C2.1 - Construct the pdf from the situation.

#### Step 1: Support and shape

The auction bounds give support $[0, 8]$. "Increases at a constant rate" gives one straight line, and the endpoint relationship gives heights $h$ and $5h$, both positive.

#### Step 2: Total area $= 1$

The full trapezoid must have area 1:

$$1 = \frac{1}{2}(8)(h + 5h) = 24h, \qquad h = \frac{1}{24}, \quad 5h = \frac{5}{24}.$$

#### Step 3: Straight-line rule

The line through $(0, 1/24)$ and $(8, 5/24)$ has slope $(5/24 - 1/24)/8 = 1/48$ and intercept $1/24 = 2/48$. Hence

#### Step 4: Complete pdf

$$f_X(x) = \begin{cases} \dfrac{x+2}{48}, & 0 \le x \le 8, \\ 0, & \text{otherwise.} \end{cases}$$

Both endpoint heights are positive, so the model is linear rather than triangular, and its trapezoidal area is 1.

### C2.2 - Calculate the tail probability $P(X > 6)$.

#### Step 5: Tail probability

The heights are $f_X(6) = 1/6$ and $f_X(8) = 5/24$. The right tail is a width-2 trapezoid:

$$P(X > 6) = \frac{1}{2}(8 - 6)\left(\frac{1}{6} + \frac{5}{24}\right) = \frac{3}{8}.$$

Thus 37.5% of modeled auctions exceed the manager's margin threshold.

### C2.3 - Calculate the interval probability $P(2 \le X \le 6)$.

#### Step 6: Interval probability

Since $f_X(2) = 1/12$ and $f_X(6) = 1/6$, the interval is a nonconstant trapezoid of width 4:

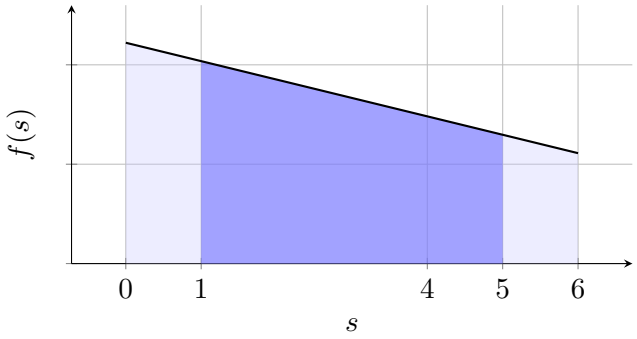
$$P(2 \le X \le 6) = \frac{1}{2}(6 - 2)\left(\frac{1}{12} + \frac{1}{6}\right) = \frac{1}{2}.$$

Thus the absorbable premium band occurs in half of the modeled auctions.



### 3.2 Solar microgrid surplus

A rural solar microgrid records the energy left after households and workshops have met their daily needs. The surplus can take any measured amount from 0 to 6 megawatt-hours. Larger surpluses become steadily less common. Among two equally narrow energy bands, one beside 0 megawatt-hours occurs twice as often as one beside 6 megawatt-hours. More than 4 megawatt-hours must be sold quickly to avoid curtailment, while the 1–5 megawatt-hour band is covered by the community battery plan.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $S$ from the narrative. Show how total area 1 determines every unknown value, give the required formula, and include 0 outside the possible range.

**C2.2** Calculate the curtailment-risk tail probability $P(S > 4)$ geometrically.

**C2.3** Calculate the central-band probability $P(1 \le S \le 5)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $S$ denote daily surplus in megawatt-hours. The narrative gives support $[0, 6]$, a straight decline, and endpoint heights $2h$ and $h$.

**Step 2: Total area** $= 1$

Normalising the full trapezoid gives

$$1 = \frac{1}{2}(6)(2h + h) = 9h, \qquad h = \frac{1}{9}.$$

**Step 3: Straight-line rule**

The line through $(0, 2/9)$ and $(6, 1/9)$ has slope $-1/54$ and intercept $2/9 = 12/54$. Thus

**Step 4: Complete pdf**

$$f_S(s) = \begin{cases} \dfrac{12 - s}{54}, & 0 \le s \le 6, \\ 0, & \text{otherwise.} \end{cases}$$

Both endpoint heights are positive and the full trapezoid has area 1.

**C2.2 - Calculate the tail probability $P(S > 4)$.**

**Step 5: Tail probability**

The endpoint heights of the right-tail trapezoid are $f_S(4) = 4/27$ and $f_S(6) = 1/9$. Therefore

$$P(S > 4) = \frac{1}{2}(6 - 4)\left(\frac{4}{27} + \frac{1}{9}\right) = \frac{7}{27} \approx 25.9\%.$$

**C2.3 - Calculate the central-band probability $P(1 \le S \le 5)$.**

**Step 6: Interval probability**

Here $f_S(1) = 11/54$ and $f_S(5) = 7/54$. The band is a width-4 trapezoid:

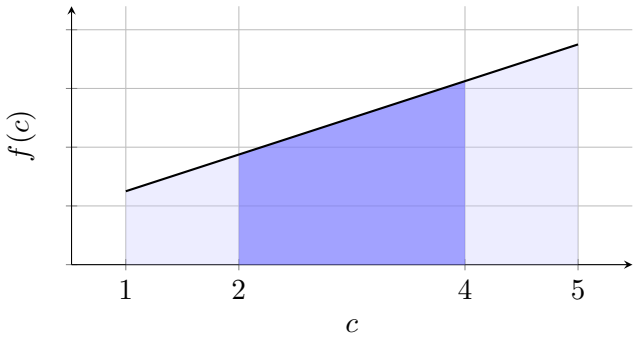
$$P(1 \le S \le 5) = \frac{1}{2}(5 - 1)\left(\frac{11}{54} + \frac{7}{54}\right) = \frac{2}{3}.$$

Thus the battery plan covers two thirds of modeled days.



## 3.3 Customs-document correction time

A freight forwarder measures how long an importer takes to correct a rejected customs document. The turnaround may be any amount from 1 to 5 days. Longer turnarounds become steadily more common because difficult cases remain open. Among two equally narrow time bands, one beside 5 days occurs three times as often as one beside 1 day. A correction taking more than 4 days risks a port penalty, while the 2–4 day band is handled by the regular compliance team.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $C$ from the narrative. Show how total area 1 determines every unknown value, give the required formula, and include 0 outside the possible range.

**C2.2** Calculate the port-penalty tail probability $P(C > 4)$ geometrically.

**C2.3** Calculate the central-band probability $P(2 \le C \le 4)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $C$ denote correction time in days. On support $[1, 5]$, the straight increase and endpoint comparison give heights $h$

and $3h$.

**Step 2: Total area $= 1$**

The complete trapezoid has area

$$1 = \frac{1}{2}(5-1)(h+3h) = 8h, \qquad h = \frac{1}{8}.$$

**Step 3: Straight-line rule**

The slope through $(1, 1/8)$ and $(5, 3/8)$ is $1/16$, so the line is $f_C(c) = (c+1)/16$. The complete pdf is

**Step 4: Complete pdf**

$$f_C(c) = \begin{cases} \dfrac{c+1}{16}, & 1 \le c \le 5, \\ 0, & \text{otherwise.} \end{cases}$$

**C2.2 - Calculate the tail probability $P(C > 4)$.**

**Step 5: Tail probability**

Since $f_C(4) = 5/16$ and $f_C(5) = 3/8 = 6/16$, the width-1 tail is

$$P(C > 4) = \frac{1}{2}(5-4)\left(\frac{5}{16} + \frac{6}{16}\right) = \frac{11}{32} = 34.375\%.$$

**C2.3 - Calculate the central-band probability $P(2 \le C \le 4)$.**

**Step 6: Interval probability**

The two heights are $f_C(2) = 3/16$ and $f_C(4) = 5/16$. Hence

$$P(2 \le C \le 4) = \frac{1}{2}(4-2)\left(\frac{3}{16} + \frac{5}{16}\right) = \frac{1}{2}.$$

The regular compliance team covers half of the modeled cases.



# 4 Triangular Distribution

**Problems:**

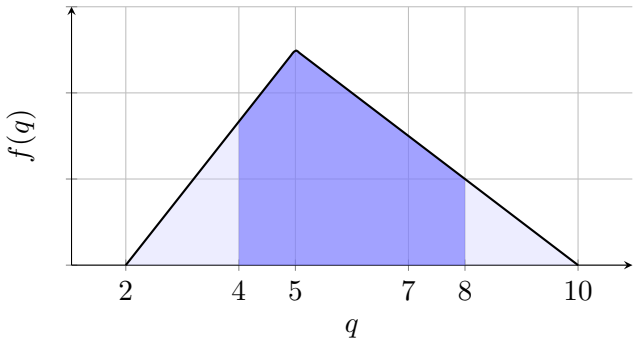

4.1. Last-mile order volume

4.2. Cold-chain produce loss

4.3. Customs clearance duration

### 4.1 Last-mile order volume

A Bogotá last-mile cooperative contracts motorcycles before a national online sales event. Next-day volume may be any amount from 2 to 10 thousand orders. Forecast records begin with no occurrence at 2 thousand, become steadily more common up to the single most likely volume of 5 thousand, and then become steadily less common until there is again no occurrence at 10 thousand. Demand above 7 thousand triggers an expensive external-courier contract, while 4–8 thousand orders is the range used to evaluate the regular fleet.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $Q$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the external-contract tail probability $P(Q > 7)$ geometrically.

**C2.3** Calculate the central-band probability $P(4 \le Q \le 8)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

The minimum, most likely value, and maximum translate to $L = 2$, $M = 5$, and $U = 10$. Rising from and falling to zero makes one triangle.

**Step 2: Total area** $= 1$

If its peak is $H$, normalisation gives

$$1 = \frac{1}{2}(10 - 2)H, \qquad H = \frac{1}{4}.$$

**Step 3: Straight-line rule**

The rising slope is $(1/4)/(5 - 2) = 1/12$; the falling slope is $-(1/4)/(10 - 5) = -1/20$. Using the zero-height endpoints,

**Step 4: Complete pdf**

$$f_Q(q) = \begin{cases} \dfrac{q-2}{12}, & 2 \le q \le 5, \\ \dfrac{10-q}{20}, & 5 < q \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

The pieces meet at height $1/4$ and the total triangular area is 1.

**C2.2 - Calculate the tail probability $P(Q > 7)$.**

**Step 5: Tail probability**

On the descending side, $f_Q(7) = 3/20$. The right tail is a triangle of base $10 - 7 = 3$ and height $3/20$:

$$P(Q > 7) = \frac{1}{2}(3)\frac{3}{20} = \frac{9}{40} = 22.5\%.$$

This is the modeled chance that external couriers are required.

**C2.3 - Calculate the interval probability $P(4 \le Q \le 8)$.**

**Step 6: Interval probability**

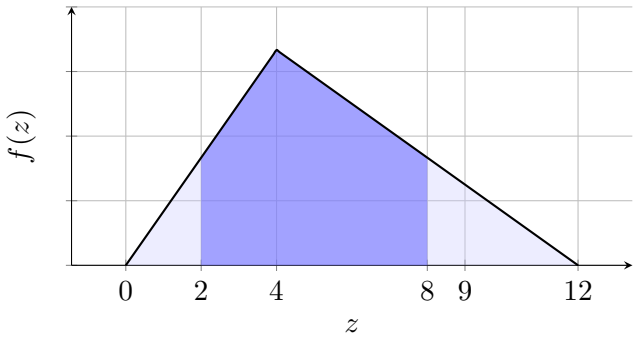
The interval crosses the mode. A clean geometric decomposition subtracts its two outside triangles. Since $f_Q(4) = 1/6$ and $f_Q(8) = 1/10$,

$$P(4 \le Q \le 8) = 1 - \underbrace{\frac{1}{2}(4-2)\frac{1}{6}}_{\text{left triangle}} - \underbrace{\frac{1}{2}(10-8)\frac{1}{10}}_{\text{right triangle}} = \frac{11}{15}.$$

Thus the regular-fleet range covers about 73.3% of modeled days.

## 4.2 Cold-chain produce loss

A produce exporter records the percentage of shipment weight lost during cooling and transport. The loss may be any amount from 0 to 12 percent. Recorded outcomes begin with no occurrence at 0 percent, become steadily more common up to the single most likely loss of 4 percent, and then become steadily less common until there is again no occurrence at 12 percent. A loss above 9 percent causes the shipment to be downgraded. The 2–8 percent band meets the exporter's normal planning assumptions.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $Z$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the downgrade-risk tail probability $P(Z > 9)$ geometrically.

**C2.3** Calculate the central-band probability $P(2 \le Z \le 8)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $Z$ denote percentage weight loss. The minimum, mode, and maximum are 0, 4, and 12.

**Step 2: Total area** $= 1$

If the triangular peak height is $H$, normalisation gives

$$1 = \frac{1}{2}(12-0)H = 6H, \qquad H = \frac{1}{6}.$$

**Step 3: Straight-line rule**

The rising side has slope $(1/6)/4 = 1/24$, while the falling side has slope $-(1/6)/8 = -1/48$. Hence

**Step 4: Complete pdf**

$$f_Z(z) = \begin{cases} \dfrac{z}{24}, & 0 \le z \le 4, \\ \dfrac{12-z}{48}, & 4 < z \le 12, \\ 0, & \text{otherwise.} \end{cases}$$

The two formulas meet at height $1/6$, and the total area is 1.

**C2.2 - Calculate the tail probability $P(Z > 9)$.**

**Step 5: Tail probability**

At $z = 9$, the falling side has height $f_Z(9) = 3/48 = 1/16$. The downgrade tail is a triangle of base $12 - 9 = 3$:

$$P(Z > 9) = \frac{1}{2}(3)\frac{1}{16} = \frac{3}{32} = 9.375\%.$$

**C2.3 - Calculate the central-band probability $P(2 \le Z \le 8)$.**

**Step 6: Interval probability**

The band crosses the mode. Since $f_Z(2) = 1/12$ and $f_Z(8) = 1/12$, subtract the two outside triangles:

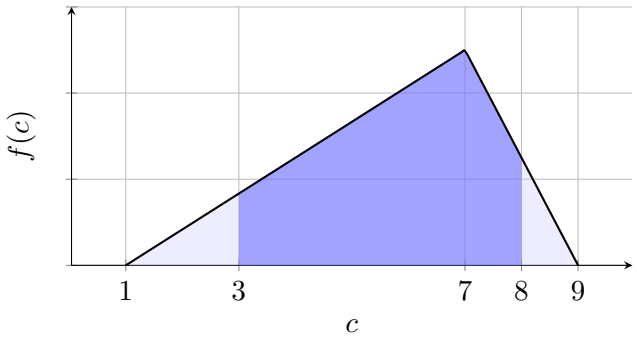
$$\begin{aligned} P(2 \le Z \le 8) &= 1 - \frac{1}{2}(2)\frac{1}{12} - \frac{1}{2}(12-8)\frac{1}{12} \\ &= 1 - \frac{1}{12} - \frac{1}{6} = \frac{3}{4}. \end{aligned}$$

Thus the normal planning band covers 75% of modeled shipments.



### 4.3 Customs clearance duration

A broker measures the time required to clear standard imported machinery. Clearance may take any amount from 1 to 9 days. Records begin with virtually no occurrence at 1 day, become steadily more common up to the single most likely duration of 7 days, and then become steadily less common until there is again no occurrence at 9 days. A duration above 8 days triggers a demurrage review, while the 3–8 day band is covered by the broker's ordinary staffing plan.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $C$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the demurrage-review tail probability $P(C > 8)$ geometrically.

**C2.3** Calculate the central-band probability $P(3 \le C \le 8)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $C$ denote clearance duration in days. The triangular support is $[1, 9]$ with mode 7.

**Step 2: Total area** $= 1$

If the peak is $H$, then

$$1 = \frac{1}{2}(9-1)H = 4H, \qquad H = \frac{1}{4}.$$

**Step 3: Straight-line rule**

The rise from $(1, 0)$ to $(7, 1/4)$ has slope $1/24$, and the fall to $(9, 0)$ has slope $-1/8$. Therefore

**Step 4: Complete pdf**

$$f_C(c) = \begin{cases} \dfrac{c-1}{24}, & 1 \le c \le 7, \\ \dfrac{9-c}{8}, & 7 < c \le 9, \\ 0, & \text{otherwise.} \end{cases}$$

**C2.2 - Calculate the tail probability $P(C > 8)$.**

**Step 5: Tail probability**

Since $f_C(8) = 1/8$, the right tail is a triangle of base 1 and height $1/8$:

$$P(C > 8) = \frac{1}{2}(9-8)\frac{1}{8} = \frac{1}{16} = 6.25\%.$$

**C2.3 - Calculate the central-band probability $P(3 \le C \le 8)$.**

**Step 6: Interval probability**

At the central band's boundaries, $f_C(3) = 1/12$ and $f_C(8) = 1/8$. Subtracting the two tail triangles gives

$$\begin{aligned} P(3 \le C \le 8) &= 1 - \frac{1}{2}(3-1)\frac{1}{12} - \frac{1}{2}(9-8)\frac{1}{8} \\ &= 1 - \frac{1}{12} - \frac{1}{16} = \frac{41}{48} \approx 85.4\%. \end{aligned}$$

# 5 Piecewise Uniform Distribution

**Problems:**

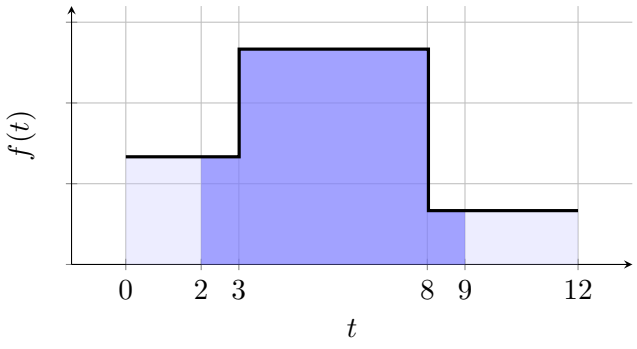

5.1. Merchant reconciliation time

5.2. Airport cargo screening time

5.3. Marketplace refund time


## 5.1 Merchant reconciliation time

A Colombian payments processor reconciles QR-code transactions before releasing merchant balances. The elapsed time can be measured to any fraction of a minute, with no automatic reconciliation taking less than 0 or more than 12 minutes. The workflow has initial validation from 0 to 3 minutes, ledger matching from 3 to 8 minutes, and exception closing from 8 to 12 minutes. Within any one stage, equal-length portions of time occur equally often. A one-minute portion during ledger matching occurs twice as often as a one-minute portion during initial validation, while a one-minute portion during exception closing occurs half as often as one during initial validation. Times above 8 minutes require the slower exception queue; the processor also evaluates a 2–9 minute service window across all three stages.



**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $T$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the exception-queue tail probability $P(T > 8)$ geometrically.

**C2.3** Calculate the central-band probability $P(2 \le T \le 9)$ geometrically.

### C2.1 - Construct the pdf from the situation.

#### Step 1: Support and shape

The stated bounds give support $[0, 12]$, with breakpoints 3 and 8. Let the initial-validation height be $k$. The verbal ratios translate the next two heights to $2k$ and $k/2$.

#### Step 2: Total area $= 1$

The three rectangles must total 1:

$$1 = 3k + (8-3)(2k) + (12-8)\frac{k}{2} = 15k, \qquad k = \frac{1}{15}.$$

#### Step 3: Straight-line rule

The density is constant on each relevant section, so no line equation is required.

Therefore

#### Step 4: Complete pdf

$$f_T(t) = \begin{cases} \frac{1}{15}, & 0 \le t < 3, \\ \frac{2}{15}, & 3 \le t < 8, \\ \frac{1}{30}, & 8 \le t \le 12, \\ 0, & \text{otherwise.} \end{cases}$$

All heights are nonnegative and the component areas sum to 1.

### C2.2 - Calculate the tail probability $P(T > 8)$.

#### Step 5: Tail probability

The exception tail is a width-4, height-1/30 rectangle:

$$P(T > 8) = (12-8)\frac{1}{30} = \frac{2}{15} \approx 13.3\%.$$

The endpoint does not affect a continuous probability.

**C2.3 - Calculate the interval probability $P(2 \le T \le 9)$.**

**Step 6: Interval probability**

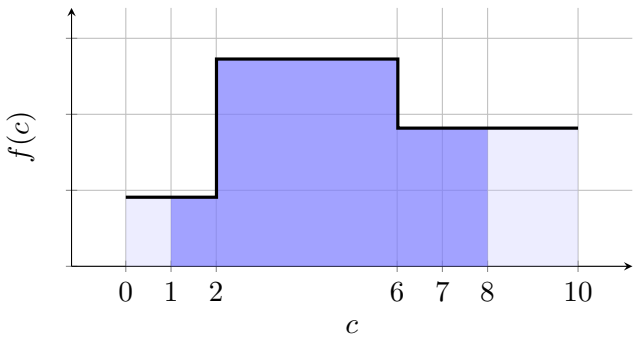
Split at both regime boundaries and add three rectangles:

$$P(2 \le T \le 9) = \underbrace{(3-2)\frac{1}{15}}_{[2,3)} + \underbrace{(8-3)\frac{2}{15}}_{[3,8)} + \underbrace{(9-8)\frac{1}{30}}_{[8,9]}$$
$$= \frac{23}{30} \approx 76.7\%.$$

This is the modeled proportion inside the processor's service window.

## 5.2 Airport cargo screening time

An airport terminal records how long export parcels spend in security screening. A parcel may take any amount from 0 to 10 hours. The terminal classifies completion times as basic review from 0 to 2 hours, secondary review from 2 to 6 hours, and specialist review from 6 to 10 hours. Within any one of these time ranges, equal-length portions occur equally often. A one-hour portion during secondary review occurs three times as often as a one-hour portion during basic review, while a one-hour portion during specialist review occurs twice as often as one during basic review. More than 7 hours risks a missed flight, and the 1–8 hour band is used for the terminal's normal staffing plan.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $C$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the missed-flight tail probability $P(C > 7)$ geometrically.

**C2.3** Calculate the central-band probability $P(1 \le C \le 8)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $C$ denote screening time in hours. The support $[0, 10]$ is divided at 2 and 6. If the first constant height is $k$, the next two heights are $3k$ and $2k$.

**Step 2: Total area** $= 1$

The three rectangles must have total area 1:

$$1 = 2k + (6-2)(3k) + (10-6)(2k) = 22k, \qquad k = \frac{1}{22}.$$

**Step 3: Straight-line rule**

The density is constant on each relevant section, so no line equation is required.

Thus

**Step 4: Complete pdf**

$$f_C(c) = \begin{cases} \frac{1}{22}, & 0 \le c < 2, \\ \frac{3}{22}, & 2 \le c < 6, \\ \frac{1}{11}, & 6 \le c \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

The component areas are 2/22, 12/22, and 8/22, which sum to 1.

**C2.2 - Calculate the tail probability** $P(C > 7)$**.**

**Step 5: Tail probability**

The missed-flight tail lies entirely in the final interval and is a rectangle:

$$P(C > 7) = (10-7)\frac{1}{11} = \frac{3}{11} \approx 27.3\%.$$

**C2.3 - Calculate the central-band probability** $P(1 \le C \le 8)$**.**

**Step 6: Interval probability**

Split the band at 2 and 6 and add three rectangular areas:

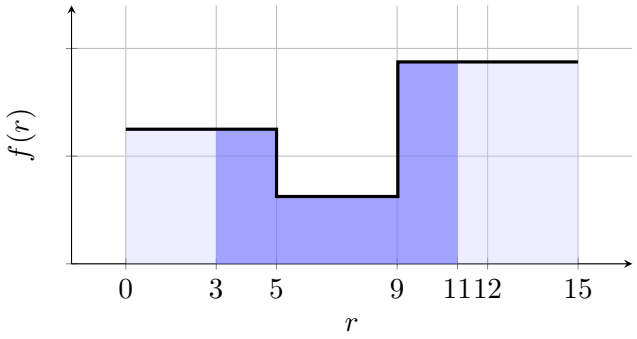
$$\begin{aligned} P(1 \le C \le 8) &= (2-1)\frac{1}{22} + (6-2)\frac{3}{22} + (8-6)\frac{1}{11} \\ &= \frac{1}{22} + \frac{12}{22} + \frac{4}{22} = \frac{17}{22} \approx 77.3\%. \end{aligned}$$



## 5.3 Marketplace refund time

An online marketplace tracks the time needed to refund returned goods after they reach a warehouse. Completion may take any amount from 0 to 15 days. The marketplace separates quick cases from 0 to 5 days, reviewed cases from 5 to 9 days, and disputed cases from 9 to 15 days. Within any one range, equal-length portions of time occur equally often. A

one-day portion among quick cases occurs twice as often as a one-day portion among reviewed cases, while a one-day portion among disputed cases occurs three times as often as one among reviewed cases. More than 12 days prompts a customer-service credit, and the 3–11 day band is used to schedule the refund team.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $R$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the service-credit tail probability $P(R > 12)$ geometrically.

**C2.3** Calculate the central-band probability $P(3 \leq R \leq 11)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $R$ denote refund time in days. The breakpoints are 5 and 9. Taking the reviewed-case height as $k$, the quick- and disputed-case heights are $2k$ and $3k$.

**Step 2: Total area** $= 1$

Normalisation of the three rectangles gives

$$1 = 5(2k) + (9-5)k + (15-9)(3k) = 32k, \qquad k = \frac{1}{32}.$$

**Step 3: Straight-line rule**

The density is constant on each relevant section, so no line equation is required.

Therefore

**Step 4: Complete pdf**

$$f_R(r) = \begin{cases} \dfrac{1}{16}, & 0 \leq r < 5, \\ \dfrac{1}{32}, & 5 \leq r < 9, \\ \dfrac{3}{32}, & 9 \leq r \leq 15, \\ 0, & \text{otherwise.} \end{cases}$$

**C2.2 - Calculate the tail probability $P(R > 12)$.**

**Step 5: Tail probability**

The service-credit tail is a width-3, height-3/32 rectangle:

$$P(R > 12) = (15 - 12)\frac{3}{32} = \frac{9}{32} = 28.125\%.$$

**C2.3 - Calculate the central-band probability $P(3 \le R \le 11)$.**

**Step 6: Interval probability**

Splitting at both changes gives

$$\begin{aligned} P(3 \le R \le 11) &= (5 - 3)\frac{1}{16} + (9 - 5)\frac{1}{32} + (11 - 9)\frac{3}{32} \\ &= \frac{4}{32} + \frac{4}{32} + \frac{6}{32} = \frac{7}{16} = 43.75\%. \end{aligned}$$



# 6 Piecewise Linear Distribution

**Problems:**

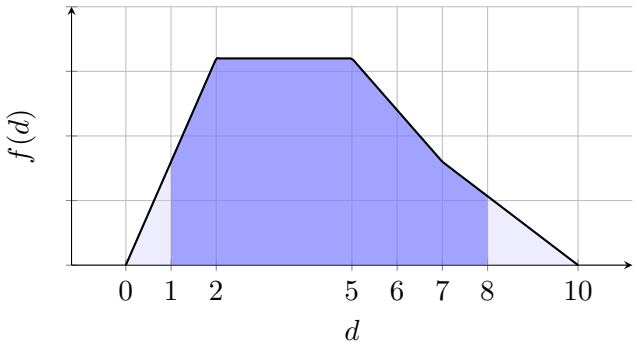

6.1. Weekly payout liquidity

6.2. Seafood export inspection time

6.3. Port truck-gate waiting time


## 6.1 Weekly payout liquidity

A Bogotá marketplace holds cash for Friday seller payouts, refunds, and chargebacks. Required liquidity may be any measured amount from 0 to 10 billion pesos. Occurrence begins at none when no cash is required, becomes steadily more common up to its highest level at 2 billion pesos, and remains equally common through 5 billion pesos. It then becomes steadily less common until requirements at 7 billion pesos occur half as often as at the highest level; from there it again becomes steadily less common until there is no occurrence at 10 billion pesos. Requirements above 6 billion pesos trigger an expensive backup credit line. The 1–8 billion peso band is used to assess how often the standard treasury plan covers the week.



**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $D$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the backup-credit tail probability $P(D > 6)$ geometrically.

**C2.3** Calculate the central-band probability $P(1 \le D \le 8)$ geometrically.

### C2.1 - Construct the pdf from the situation.

#### Step 1: Support and shape

The story gives support $[0, 10]$ and breakpoints $2, 5, 7$. If the maximum is $H$, the corresponding heights are $0, H, H, H/2, 0$. Thus the geometry is a base-2 triangle, width-3 rectangle, width-2 trapezoid, and base-3 triangle.

#### Step 2: Total area $= 1$

Normalisation gives

$$1 = \frac{1}{2}(2)H + 3H + \frac{1}{2}(2)\left(H + \frac{H}{2}\right) + \frac{1}{2}(3)\frac{H}{2} = \frac{25}{4}H, \qquad H = \frac{4}{25}.$$

#### Step 3: Straight-line rule

Deriving each line from its two endpoint values gives slopes $2/25$, $0$, $-1/25$, and $-2/75$, respectively. Therefore

#### Step 4: Complete pdf

$$f_D(d) = \begin{cases} \dfrac{2d}{25}, & 0 \le d \le 2, \\ \dfrac{4}{25}, & 2 < d \le 5, \\ \dfrac{9-d}{25}, & 5 < d \le 7, \\ \dfrac{2(10-d)}{75}, & 7 < d \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

Adjacent equations agree at every breakpoint, and the geometric areas total 1.

### C2.2 - Calculate the tail probability $P(D > 6)$.

#### Step 5: Tail probability

The tail begins within the declining 5–7 segment. Its heights at 6 and 7 are $3/25$ and $2/25$. From 6 to 7 is a trapezoid; from 7 to 10 is a triangle:

$$P(D > 6) = \underbrace{\frac{1}{2}(7-6)\left(\frac{3}{25} + \frac{2}{25}\right)}_{\text{trapezoid}} + \underbrace{\frac{1}{2}(10-7)\frac{2}{25}}_{\text{triangle}}$$
$$= \frac{1}{10} + \frac{3}{25} = \frac{11}{50} = 22\%.$$

This is the modeled frequency with which backup credit is triggered.

**C2.3 - Calculate the interval probability $P(1 \le D \le 8)$.**

**Step 6: Interval probability**

Although the interval spans every structural piece, its complement consists of two simple tail triangles. Since $f_D(1) = 2/25$ and $f_D(8) = 4/75$,

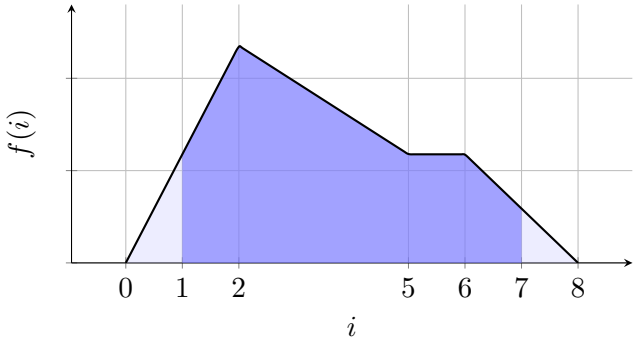
$$P(1 \le D \le 8) = 1 - \underbrace{\frac{1}{2}(1)\frac{2}{25}}_{[0,1]} - \underbrace{\frac{1}{2}(10-8)\frac{4}{75}}_{[8,10]}$$
$$= 1 - \frac{1}{25} - \frac{4}{75} = \frac{68}{75} \approx 90.7\%.$$

Thus the standard treasury range covers about 90.7% of modeled weeks.



## 6.2 Seafood export inspection time

A coastal inspection office records how long chilled seafood shipments wait for export clearance. A shipment may wait any amount from 0 to 8 hours. Occurrence begins at none for immediate clearance, becomes steadily more common up to its highest level at 2 hours, and then becomes steadily less common until a 5-hour wait occurs half as often as a 2-hour wait. That level remains unchanged through 6 hours, after which occurrence becomes steadily less common until there is none at 8 hours. A wait above 6 hours requires an overnight cold-storage booking, while the 1–7 hour band is covered by the ordinary inspection shift.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $I$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the overnight-storage tail probability $P(I > 6)$ geometrically.

**C2.3** Calculate the central-band probability $P(1 \le I \le 7)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $I$ denote inspection time in hours. The support is $[0, 8]$, with heights $0, H, H/2, H/2, 0$ at $0, 2, 5, 6, 8$.

**Step 2: Total area** $= 1$

The triangle, trapezoid, rectangle, and final triangle must have total area 1:

$$\begin{aligned} 1 &= \frac{1}{2}(2)H + \frac{1}{2}(5-2)\left(H + \frac{H}{2}\right) + (6-5)\frac{H}{2} + \frac{1}{2}(8-6)\frac{H}{2} \\ &= \frac{17}{4}H, \qquad H = \frac{4}{17}. \end{aligned}$$

**Step 3: Straight-line rule**

The changing segments have slopes $2/17$, $-2/51$, and $-1/17$; the constant segment needs no line equation. Using the appropriate endpoints in point-slope form gives the formulas below.

**Step 4: Complete pdf**

$$f_I(i) = \begin{cases} \dfrac{2i}{17}, & 0 \le i \le 2, \\ \dfrac{16-2i}{51}, & 2 < i \le 5, \\ \dfrac{2}{17}, & 5 < i \le 6, \\ \dfrac{8-i}{17}, & 6 < i \le 8, \\ 0, & \text{otherwise.} \end{cases}$$

Adjacent formulas agree at 2, 5, and 6, and their areas sum to 1.

**C2.2 - Calculate the tail probability** $P(I > 6)$**.**

**Step 5: Tail probability**

The overnight-storage tail is the final triangle, with base $8 - 6 = 2$ and height $2/17$:

$$P(I > 6) = \frac{1}{2}(2)\frac{2}{17} = \frac{2}{17} \approx 11.8\%.$$

**C2.3 - Calculate the central-band probability** $P(1 \le I \le 7)$**.**

**Step 6: Interval probability**

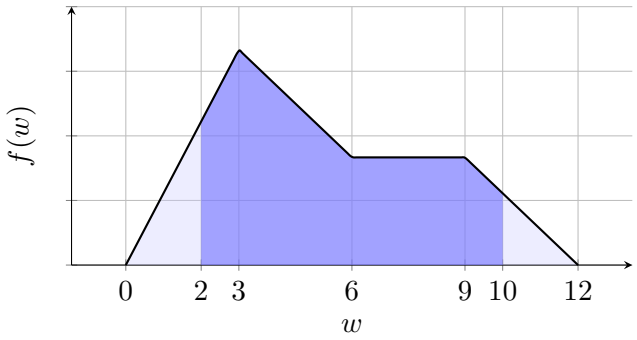
The complement contains two triangles. Their boundary heights are $f_I(1) = 2/17$ and $f_I(7) = 1/17$, so

$$\begin{aligned} P(1 \le I \le 7) &= 1 - \frac{1}{2}(1)\frac{2}{17} - \frac{1}{2}(8-7)\frac{1}{17} \\ &= 1 - \frac{1}{17} - \frac{1}{34} = \frac{31}{34} \approx 91.2\%. \end{aligned}$$



### 6.3 Port truck-gate waiting time

A port terminal records how long trucks wait between joining the entry queue and reaching the inspection gate. A wait may be any amount from 0 to 12 hours. Occurrence begins at none for no wait, becomes steadily more common up to its highest level at 3 hours, and then becomes steadily less common until a 6-hour wait occurs half as often as a 3-hour wait. That lower level remains unchanged through 9 hours, after which occurrence becomes steadily less common until there is none at 12 hours. A wait above 10 hours causes a missed loading slot, while the 2–10 hour band is used for routine gate staffing.




**Questions/Tasks.**

**C2.1** Construct the complete probability density function of $W$ from the narrative. Show how total area 1 determines every unknown value, give all required formulas, and include 0 outside the possible range.

**C2.2** Calculate the missed-slot tail probability $P(W > 10)$ geometrically.

**C2.3** Calculate the central-band probability $P(2 \le W \le 10)$ geometrically.

**C2.1 - Construct the pdf from the situation.**

**Step 1: Support and shape**

Let $W$ denote gate waiting time in hours. The support is $[0, 12]$, and the heights at $0, 3, 6, 9, 12$ are $0, H, H/2, H/2, 0$.

**Step 2: Total area** $= 1$

Normalisation gives

$$1 = \frac{1}{2}(3)H + \frac{1}{2}(6-3)\left(H + \frac{H}{2}\right) + (9-6)\frac{H}{2} + \frac{1}{2}(12-9)\frac{H}{2}$$
$$= 6H, \qquad H = \frac{1}{6}.$$

**Step 3: Straight-line rule**

The successive slopes are $1/18$, $-1/36$, $0$, and $-1/36$. Using the endpoint values gives the formulas shown below.

**Step 4: Complete pdf**

The complete pdf is therefore

$$f_W(w) = \begin{cases} \dfrac{w}{18}, & 0 \le w \le 3, \\ \dfrac{9-w}{36}, & 3 < w \le 6, \\ \dfrac{1}{12}, & 6 < w \le 9, \\ \dfrac{12-w}{36}, & 9 < w \le 12, \\ 0, & \text{otherwise.} \end{cases}$$

The formulas meet at the stated heights and the geometric areas total 1.

**C2.2 - Calculate the tail probability $P(W > 10)$.**

**Step 5: Tail probability**

At $w = 10$, the final side has height $f_W(10) = 2/36 = 1/18$. The right tail is a triangle of base 2:

$$P(W > 10) = \frac{1}{2}(12 - 10)\frac{1}{18} = \frac{1}{18} \approx 5.6\%.$$

**C2.3 - Calculate the central-band probability $P(2 \le W \le 10)$.**

**Step 6: Interval probability**

The left outside triangle has base 2 and height $f_W(2) = 1/9$; the right outside triangle is the tail just calculated. Thus

$$P(2 \le W \le 10) = 1 - \frac{1}{2}(2)\frac{1}{9} - \frac{1}{18} = 1 - \frac{1}{9} - \frac{1}{18} = \frac{5}{6}.$$

Thus routine staffing covers about 83.3% of modeled waits.

